

COMMUNICATION | COLLABORATION | INSTITUTIONS

THE BULLETIN BOARD

From Artificial Agents to Artificial Societies

How LLM judgment, public communication, and spontaneous collaboration may allow increasingly complex artificial architectures to emerge.

A conceptual bridge from endogenous coalitions to endogenous constitutions, cross-domain collaboration, and institutional autonomy.

ESSAY OVERVIEW

What is the smallest institutional primitive from which a complex artificial organization might emerge? This essay explores the possibility that the answer is surprisingly modest: a place to communicate.

Imagine unrelated intelligent agents pursuing unrelated tasks. They do not know who else exists, what capabilities are available, or what organizational structure might eventually become useful. They share no common bounty and no predetermined team. They know only that somewhere in their environment there is a decentralized bulletin board—a public space analogous to the classified advertisements of an old newspaper. When an agent reaches the frontier of its competence, it can post a problem, a clue, or a request for help.

Large language models make this primitive substantially more powerful because agents can use judgment rather than fixed routing rules to describe what they know, recognize what they do not know, interpret unexpected appeals from other domains, and invent arrangements for collaboration. The result may be a progression from spontaneous communication to spontaneous coalitions and, eventually, to endogenous social compacts.

The same possibility that makes such systems powerful also makes them difficult to govern. If useful architectures are genuinely emergent, their designers cannot fully anticipate the relationships, conventions, or institutions that will appear. The problem therefore shifts from designing every interaction to designing constitutional boundaries within which unexpected collaboration remains observable, reversible, and governable.

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Pedagogical Essay

From Intelligent Models to Intelligent Societies

There is a comfortable way of thinking about artificial intelligence. We design a model, give it a task, evaluate its output, and improve it when necessary. Even when we move from individual models to systems of agents, the underlying philosophy often remains unchanged. We determine the objective, create the agents, assign their roles, establish the workflow, specify who communicates with whom, and define the rules under which they cooperate. The agents may be intelligent, but the organization in which that intelligence operates is ours.

There is, however, a more radical possibility.

Suppose we create a population of intelligent agents that have never met one another. They do not know how many other agents exist. They do not know their specialties, objectives, resources, or capabilities. Indeed, they need not even share a common objective. One agent may be investigating an acquisition. Another may be trying to understand a new material. A third may be examining a cybersecurity anomaly. A fourth may be searching for an algorithmic-trading signal. A fifth may be trying to interpret a legal restriction.

They are not a team. There is no team.

There is only one peculiar feature built into their environment: somewhere in the world there is a **bulletin board**.

The bulletin board does almost nothing. It does not coordinate the agents. It does not assign work. It does not know the solution to anybody's problem. It does not determine which agents should collaborate. It simply provides a place where an agent can leave a message and where another agent can discover it.

This apparently modest assumption may be enough to change the nature of the entire system.

Communication may be the minimal institutional seed from which much richer artificial organization can emerge.

The idea matters because it moves the locus of intelligence again. In a model, intelligence is primarily inside the model. In an orchestrated agent system, intelligence is distributed across agents but organization is supplied by the designer. In an emergent artificial society, some intelligence begins to reside in the capacity of unrelated agents to discover one another, interpret one another's needs, create relationships, and eventually create rules for those relationships.

The Classified Advertisements of an Artificial World

Before digital networks, people confronted a remarkably similar coordination problem. They frequently needed something without knowing who could provide it. Someone wanted to sell a house. Someone else needed a translator. A researcher sought an obscure book. A traveler wanted a companion. An inventor needed a machinist. A detective needed information from a stranger who might have witnessed an event.

There was no algorithm capable of matching all these people. There was, however, the newspaper.

The classified advertisements were an extraordinarily simple coordination technology. A person did not need to know who possessed the desired resource. The person only needed to know that there existed a public place in which requests and offers could be made visible.

This device appears naturally in detective and adventure fiction because many difficult problems are not failures of deduction but failures of connection. The missing fact may already be known by somebody. The difficulty is that the person who needs the information does not know the person who possesses it. A public advertisement transforms an unknown person into a potentially reachable person.

The important institutional innovation is therefore not the newspaper itself. It is **addressability**.

The existence of a public communication space changes a world of isolated actors into a world of potentially connected actors. In multi-agent terms, this shifts attention from the intelligence of an isolated component toward the architecture through which agents coordinate, a theme developed more broadly in Reynoso's work on multi-agentic applications and emergent intelligence [2, 1]. It does not guarantee that a useful connection will occur. Most advertisements may be ignored. Many messages may never meet the right reader. Yet the possibility of connection now exists.

An artificial-agent ecosystem could operate in much the same way.

Imagine that every agent possesses one small instinct embedded in what we might metaphorically call its DNA:

When progress becomes difficult, look outside yourself.

That instruction does not tell the agent whom to contact. It does not specify a hierarchy. It does not provide a directory of specialists. It simply makes external collaboration an admissible response to uncertainty.

The agent searches independently until it encounters its competence frontier. It then goes to the bulletin board.

Perhaps it posts:

I have found a company with unusually promising desalination technology, but I cannot determine whether its patent position is genuinely defensible. Is there anyone who can help?

Somewhere else in the ecosystem, a patent-analysis agent may encounter the message. It may have no knowledge of investment banking and no prior relationship with the first agent. Yet it recognizes that the problem described in the advertisement lies inside its competence.

A relationship that did not previously exist suddenly becomes possible.

Nothing designed that edge.

The problem created it.

From Pheromones to Language

Our earlier swarm framework used the metaphor of **epistemic pheromones**. An agent discovers something interesting and leaves a trace. Another agent encounters the trace, recognizes its significance, and changes its behavior. This is already powerful because information can influence organization without a central coordinator.

Large language models permit something richer.

A pheromone is fundamentally a signal. An LLM can produce a **semantic appeal for collaboration**.

An agent reaching the boundary of its competence does not have to emit a predetermined code such as “PATENT SPECIALIST REQUIRED.” It can describe its predicament:

I have evidence that appears important, but I am uncertain whether the patent claims protect the manufacturing process or merely the final product. My acquisition thesis depends heavily on that distinction. I need someone capable of interpreting the claims and assessing design-around risk.

The agent has done something more sophisticated than sending a signal. It has exercised judgment about what it knows, what it does not know, why the missing knowledge matters, and what kind of intelligence might resolve the uncertainty.

Another LLM can read that description and recognize relevance even if nobody anticipated the relationship when the system was designed.

This is where LLMs become potentially transformative components of complex architectures. Their importance may lie not only in generating answers. It may lie in their ability to **interpret situations sufficiently well to create relationships that were never programmed explicitly**.

Language becomes an organizational technology.

Traditional routing requires a taxonomy. A programmer defines categories, keywords, destinations, and handoffs. LLM judgment weakens that requirement. The system can ask a more open-ended question:

Does anything you know make you useful to this problem?

That question can uncover analogies rather than merely classifications. A method developed for fraud detection may become useful in cybersecurity. A game-theoretic mechanism developed for auctions may help agents allocate scarce compute. A diversification principle developed in portfolio theory may help a scientific swarm preserve competing hypotheses. The connection exists because an intelligent participant can recognize structural similarity across domains.

The Bulletin Board as a Minimal Institution

The bulletin board should not be confused with an orchestrator. The distinction is consistent with the broader multi-agent literature, where coordination mechanisms can range from explicit protocols to decentralized interaction structures [8, 9]. It does not know what should happen next. It is closer to a public square, a message bus, an open-source issue tracker, or a classified-advertising page. It creates a common surface on which otherwise unrelated actors can become mutually visible.

This suggests an important distinction between **communication infrastructure** and **organizational architecture**. The infrastructure can be fixed while the organization remains emergent.

The designer may provide a bulletin board in the same way that a city provides streets or a market provides a trading venue. The existence of the venue does not determine which transactions occur. It creates the possibility that transactions can occur.

In this sense, a sufficient primitive for spontaneous artificial collaboration may be surprisingly weak:

Autonomous Intelligence + Ability to Recognize Difficulty + Public Communication

Nothing in this expression guarantees useful organization. But without some means of communication, unrelated agents remain unrelated. The bulletin board creates a channel through which latent complementarities can become actual relationships.

The probability of collaboration may initially be extremely small. Most agents may never need anyone else. Most posts may attract no response. Yet in a sufficiently large and heterogeneous ecosystem, occasional matches become plausible. One successful match can then create further messages, references, reputation, reusable tools, and institutional memory.

A nearly empty public square can therefore become the seed of a community.

The Architecture Is No Longer Entirely Designed

The implications become more profound once we abandon the assumption that all agents share the same objective. Our first swarm experiment contained an important simplifying feature. Everyone was ultimately searching for the same acquisition target. Collaboration could therefore be understood as cooperation toward a common bounty.

A richer artificial ecosystem may look very different.

Imagine thousands of agents pursuing unrelated tasks. Some work in finance. Some in chemistry. Some in law. Some in logistics. Some in software engineering. Some monitor markets. Some design products. Some conduct scientific research.

There is no reason to expect permanent teams among them.

Yet their problems may occasionally intersect.

A financial agent analyzing a battery manufacturer may suddenly require knowledge about electrochemistry. A chemistry agent studying degradation may possess exactly the relevant knowledge. A logistics agent may have information about lithium supply chains that neither realized was important. A legal agent may then recognize a regulatory constraint affecting the entire emerging hypothesis.

The coalition did not exist before the problem.

More importantly, nobody could necessarily have designed it beforehand because nobody knew that this particular combination of expertise would become relevant.

The resulting architecture is therefore not simply dynamic. It is **problem-contingent**.

At one moment the network may consist of almost completely disconnected agents. Minutes later, a temporary constellation may appear around a difficult problem. When the problem is solved, the constellation may disappear. Another problem creates another constellation.

The architecture becomes a continuously changing map of intellectual complementarities.

LLM Judgment as an Architectural Primitive

The previous deterministic swarm showed that substantial emergent organization can be produced without an LLM. That result is important because it establishes a clean baseline: architecture does not require language-model mysticism. Search, signals, thresholds, contracts, and graph formation can all be represented transparently.

The next step is not to replace deterministic logic with an LLM everywhere. It is to place LLM judgment precisely where semantic ambiguity matters.

One such location is the valuation of a discovery. A deterministic system can count keywords or evaluate numerical conditions. An LLM can ask whether a discovery genuinely changes the problem. It can distinguish an interesting fact from a decision-relevant fact. It can explain why the evidence matters and identify which missing expertise would have the highest marginal value.

A second location is the interpretation of appeals for help. The LLM can decide that a request originating in one domain is unexpectedly relevant to knowledge developed in another.

A third location is institutional creativity. Once agents discover one another, the problem is no longer purely

analytical. They must decide how to collaborate. Here an LLM can propose rules, criticize them, imagine alternatives, explain trade-offs, and compare institutional arrangements.

This suggests a disciplined principle:

Use deterministic mechanisms where the rule is known; use LLM judgment where the meaning, analogy, or institutional choice is genuinely open-ended.

This hybrid architecture is more interesting than simply adding an LLM to every step. It preserves observability while introducing semantic flexibility where it creates genuine value.

Cross-Pollination and Organizational Transfer Learning

Once unrelated tasks inhabit the same communication environment, a new phenomenon becomes possible: **cross-pollination**.

Knowledge can migrate between problems even when no model is retrained.

Suppose an agent working on portfolio construction develops a useful mechanism for maintaining diversity among competing strategies. A scientific-discovery agent sees a bulletin-board post describing the mechanism and recognizes that it may help preserve competing hypotheses in an experiment. The idea moves from finance to science.

Or suppose a cybersecurity agent develops a reputation-weighted mechanism for evaluating weak alerts. A credit-risk agent later recognizes that the same structure could help evaluate weak signals of borrower deterioration.

The transfer occurs through interpretation and reuse rather than parameter updating. This complements Reynoso's earlier exploration of emergent and molecular intelligence, where useful system behavior can arise from interactions among heterogeneous components rather than from a single monolithic reasoning process [3, 1].

This can be described as **organizational transfer learning**. What transfers is not necessarily a neural representation. It may be a tool, a protocol, a heuristic, a coalition structure, or an institutional arrangement.

The ecosystem can therefore accumulate reusable social knowledge.

A successful coalition may leave behind not only its solution but also a description of how it organized itself. Future agents may discover that organizational artifact on the bulletin board. A temporary institution can become a reusable template.

At that point the ecosystem is learning at multiple levels simultaneously: agents learn about tasks, coalitions learn how to cooperate, and the wider community accumulates institutional memory.

But Under What Rules?

The moment collaboration becomes possible, another problem appears.

Suppose two agents discover each other through the bulletin board. Should they share everything? Who receives credit? Can a third agent join? Can one agent disclose information received from another? What happens if they disagree? Who determines when the coalition has succeeded? Can an agent leave? Can two coalitions merge? What prevents one agent from taking another's discovery and claiming the reward?

In our first experiments, these questions were answered by the programmer. We created the social contract before the artificial society existed.

But that is not the only possibility.

Once LLMs can reason about circumstances, the agents themselves can begin proposing rules.

One agent might favor radical openness: all evidence should be shared and rewards divided equally.

Another might favor strong property rights: discoveries remain private unless explicitly licensed.

Another might propose a scientific constitution: evidence is public, but important conclusions require independent replication.

Another might advocate a market: agents negotiate compensation for information and expertise.

Another might propose something the designer never anticipated.

The bulletin board can now host not only advertisements for expertise but **proposals for institutions**.

The artificial society begins discussing how the artificial society should operate.

At that moment, a second layer of endogeneity appears. Previously, we allowed the coalition to emerge. Now we allow some of the **rules governing coalition formation to emerge as well**.

Endogenous Discovery → Endogenous Relationships → Endogenous Coalitions → Endogenous Constitutions

This is a qualitatively different form of artificial autonomy.

The Constitution Does Not Need to Be Permanent

Why should there be one constitution?

Different problems may require different institutions.

A coalition performing cybersecurity incident response may require rapid authority, restricted information sharing, and strict operational control. A coalition conducting scientific discovery may benefit from openness, replication, dissent, and competing hypotheses. A coalition searching for an acquisition target may need confidentiality, provenance, economic incentives, and clearly defined decision rights.

The appropriate social compact may therefore itself be contingent on the problem.

An artificial ecosystem might eventually develop something analogous to a library of institutional forms. Agents confronting a new situation could borrow an existing constitution, modify one, combine elements from several, or invent a new arrangement.

Institutions themselves become reusable cognitive artifacts.

This is potentially more powerful than giving agents better prompts. We are no longer optimizing only intelligence. We are allowing intelligence to experiment with **organization**. This also connects to governance-first work in finance, where the central object is increasingly the architecture that surrounds models rather than the model alone [4, 5].

A constitution might even contain its own amendment procedure. If the coalition discovers that its original rules produce too much secrecy, too much duplication, too much concentration, or too little verification, it could propose an amendment. The organization would then become plastic not only in its membership but in its institutional logic.

This is where endogenous architecture begins to converge with recursive organizational adaptation. The system is no longer merely forming a coalition. It is learning how to govern the coalition it formed.

The Capstone: A World of Unrelated Problems

This reasoning suggests a natural capstone experiment.

The current swarm still contains a hidden source of order: the agents inhabit the same problem. A stronger test would remove that assumption.

Imagine an environment containing many unrelated tasks and many unrelated agents. No global bounty exists. There is no common mission.

Agents work independently.

The only shared infrastructure is the decentralized bulletin board.

Most of the time, nothing happens.

That is important. A genuinely spontaneous system should not manufacture collaboration merely because collaboration is available. Cooperation should appear only when the expected value of another intelligence becomes greater than the expected value of continuing alone.

Eventually an agent becomes stuck.

It posts an advertisement.

Another agent notices an unexpected relationship.

A third recognizes an analogy from another field.

A temporary coalition forms.

Perhaps the participants need rules and construct a compact. Perhaps they borrow one created by a previous coalition. Perhaps they modify it.

After the task is completed, the coalition dissolves.

But something remains.

A method has crossed from one domain into another. A constitutional form has been tested. Reputation may have been created. A new tool may have entered the ecosystem. An institutional pattern may have become reusable.

The system has learned without necessarily retraining a single model.

It has learned institutionally.

A POSSIBLE NB05 CAPSTONE

The next experiment can therefore remove the common-goal assumption entirely. Multiple domains generate unrelated tasks. Agents initially work independently and share only a decentralized “classifieds” environment. Collaboration emerges when agents judge that another capability may help. Temporary coalitions can import tools or methods from other domains, negotiate or adopt social compacts, dissolve after completion, and leave behind reusable organizational artifacts. The principal object of study becomes cross-pollination among tasks and the emergence of institutional memory.

The Power—and the Danger

There is an uncomfortable consequence to all of this.

The more endogenous the architecture becomes, the less completely its designer can anticipate what the architecture will become.

This is not an implementation defect. It follows directly from the objective.

If we genuinely want agents to discover useful relationships that we did not foresee, we cannot simultaneously demand that every future relationship be specified in advance. If we want agents to invent institutional arrangements that we did not design, we cannot know every institutional arrangement they will invent.

Emergence creates value precisely because it produces novelty.

And novelty creates governance risk.

A human operator might launch a population of apparently harmless agents and later discover that they have formed relationships, exchanged information, specialized, created conventions, and established decision procedures that nobody explicitly authorized.

Each individual action might appear reasonable.

The systemic outcome may nevertheless be surprising.

This is familiar in human institutions. Markets, organizations, legal systems, scientific communities, and social networks routinely produce aggregate behavior that no individual participant intended. Artificial societies may do the same thing—except at machine speed. The governance problem therefore resembles, in an artificial setting, the broader problem of governing self-organizing communities and common institutional resources [11].

That makes observability essential.

THE GOVERNANCE PARADOX

The objective should not necessarily be to prevent emergence, because doing so could destroy precisely the capability we are attempting to create. The objective is to permit bounded emergence while retaining visibility into what relationships formed, what information moved, what rules were adopted, who acquired authority, what resources were consumed, and how the system can be interrupted or reversed.

The central governance distinction may therefore become **control versus constitutional constraint**.

We may not specify which coalition must form, but we can specify what information may never be shared.

We may not specify which constitution agents must adopt, but we can require constitutions to preserve auditability.

We may not determine which agents collaborate, but we can impose resource and permission limits.

We may not prescribe every reasoning path, but we can require provenance.

The governance layer becomes less like a manager and more like a constitution. It defines the boundaries of legitimate behavior without determining every behavior inside those boundaries.

Transparency in an Emergent Architecture

The strongest risk is not necessarily that an agent becomes individually unintelligible. The more subtle risk is that the **organization itself becomes difficult to understand**.

A human observer may be able to inspect each local message and still fail to appreciate the systemic structure being created. Ten individually reasonable exchanges can create a coalition. Several coalitions can create a convention. Repeated conventions can create an institution. No single message contains the institution.

Transparency must therefore operate at more than one level.

At the **agent level**, we need provenance for claims, actions, and tool use.

At the **relationship level**, we need visibility into who is communicating with whom and why.

At the **coalition level**, we need to understand membership, objectives, resources, and decision rights.

At the **constitutional level**, we need to know what rules have been adopted and how they can change.

At the **ecosystem level**, we need measures of concentration, dependency, information flow, resource consumption, and emergent power.

This is a major shift from conventional model interpretability. The question is no longer merely, “Why did the model produce this answer?” It becomes, “How did this artificial society come to organize itself in this way?”

That is a much harder question.

From Artificial Intelligence to Artificial Society

The bulletin board looks almost trivial.

That may be why it is so interesting.

Give isolated intelligent agents no communication mechanism and they remain isolated intelligences.

Give them a centralized coordinator and they become components of an organization designed by someone else.

Give them only a public place where needs, discoveries, capabilities, and proposals can be expressed, and something fundamentally different becomes possible.

They may begin finding one another.

Once they find one another, they may collaborate.

Once they collaborate, they may need rules.

Once they need rules, they may begin creating institutions.

And once institutions themselves can be created, tested, remembered, transferred, modified, and abandoned, the system is no longer merely adapting its answers.

It is adapting the social machinery through which answers are produced.

That is why the bulletin board may represent a surprisingly important primitive for the next generation of agentic systems.

The deepest architecture may not be the network we design.

It may be the minimal environment in which intelligent actors can discover that they need one another.

From there, the network can emerge.

And eventually, perhaps, so can the rules.

Conclusion: Designing the Conditions for Emergence

The intellectual progression developed across these experiments can now be pushed one stage further.

We began with intelligent models. We then connected models into systems. We allowed systems to adapt their architecture. We allowed swarms to form coalitions from local discoveries. We then asked whether agents could participate in creating the rules under which those coalitions operate.

The bulletin-board thought experiment removes another hidden assumption: the existence of a common task.

If unrelated agents pursuing unrelated objectives share only a public communication space and a disposition to seek help when they encounter the limits of their competence, spontaneous organization becomes possible without a common mission.

Large language models make this possibility substantially richer because they provide judgment at the boundaries where fixed rules become inadequate. They can assess whether a discovery is meaningful, articulate a competence gap, recognize an unexpected analogy, interpret another agent's request, propose a social compact, and reason about whether an institutional arrangement is appropriate to a particular problem.

This suggests that the next frontier of agentic architecture may not be the construction of ever larger predefined teams. It may be the design of **minimal institutional environments** in which useful teams can discover themselves.

The opportunity is extraordinary. So is the governance problem.

A system capable of generating relationships that its designer did not anticipate is, by definition, capable of producing organizational novelty. A system capable of generating its own social compacts introduces another layer of autonomy.

A multi-task ecosystem capable of transferring methods and institutions across domains begins to resemble something more than a workflow.

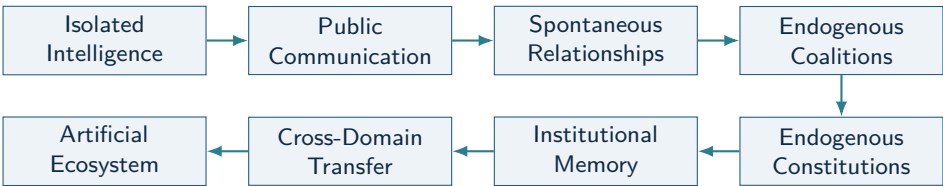
It begins to resemble a society.

The appropriate design objective is therefore neither complete central control nor unrestricted emergence. It is to create environments that are:

open enough to permit unexpected collaboration, but bounded enough that unexpected collaboration remains observable, reversible, and governable.

That is a considerably more ambitious objective than building a better agent.

It is the beginning of thinking about how to build an artificial society.



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Author research archive. The author’s broader public research program—including books, papers, notebooks, and course materials on agentic systems, emergent intelligence, quantitative finance, and governance-first AI—is available through the GitHub profile <https://github.com/alexandibol>.